

Connectivity properties of two-dimensional fracture networks with stochastic fractal correlation

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[1] We present a theoretical and numerical study of the connectivity of fracture networks with fractal correlations. In addition to length distribution, this spatial property observed on most fracture networks conveys long-range correlation that may be crucial on network connectivity. We especially focus on the model that comes out relevant to natural fracture network: a fractal density distribution for the fracture centers (dimension D) and a power law distribution for the fracture lengths (exponent a , $n(l) \sim l^{-a}$). Three different regimes of connectivity are identified depending on D and on a . For $a < D + 1$ the length distribution prevails against fractal correlation on network connectivity. In this case, global connectivity is mostly ruled by the largest fractures whose length is of the order of the system size, and thus the connectivity increases with scale. For $a > D + 1$ the connectivity is ruled by fractures much smaller than the system size with a strong control of spatial correlation; the connectivity now decreases with system scale. Finally, for the self-similar case ($a = D + 1$), which corresponds to the transition between the two previous regimes, connectivity properties are scale invariant: percolation threshold corresponds to a critical fractal density and the average number of intersections per fracture at threshold is a scale invariant as well. **INDEX TERMS:** 1829 Hydrology: Groundwater hydrology; 1869 Hydrology: Stochastic processes; 8010 Structural Geology: Fractures and faults; 3250 Mathematical Geophysics: Fractals and multifractals; **KEYWORDS:** fractured media, connectivity, scaling law, fractal, correlation, self-similarity

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1. Introduction

[2] The connectivity of fractures, embedded in impermeable or low permeability geological formations, is a critical feature controlling fluid and chemical movement in these systems [Long and Billaux, 1987; Margolin *et al.*, 1998]. The relationship between fracture-network connectivity and fracture geometrical properties has been intensively studied for the fracture length distribution [Berkowitz, 1995; Bour and Davy, 1997, 1998; Renshaw, 1999], and for the distribution of fracture orientations [Balberg *et al.*, 1984, 1991; Robinson, 1983; Robinson, 1984]. We lay a special emphasis on the recent results from Bour and Davy [1997, 1998] and de Dreuzy *et al.* [2000] which show that percolation theory does not necessarily apply on fracture networks whose length distribution is a power law. This length distribution model, which is characteristic of natural fracture systems [Bonnet *et al.*, 2001, and references therein], emphasizes the unlimited nature of fracture sizes which are observed from microns to 100's of kilometers. Associated with a random distribution of fracture positions, it provides a rich variety of behavior for connectivity properties, from classical percolation theory to superposition of

infinite fractures (i.e., fractures of length larger than the system size) [Bour and Davy, 1998, 1997; de Dreuzy *et al.* 2000, 2001a, 2001b]. The nature of the connectivity model is fixed by the power law length exponent a . For instance for 2-D networks, if $a > 3$, connectivity is ruled by the smallest fractures, and the classical rules of percolation theory apply; if $a < 2$, the role of small fractures is negligible, and in terms of connectivity, the system is similar to the superposition of infinite fractures [Snow, 1969; Kiraly, 1975]; if $2 < a < 3$, both large and small fractures rule the connectivity, which is significantly scale dependent in contrast to percolation theory.

[3] All the aforementioned studies refer to a uniform stochastic distribution of fracture positions. This assumption is somehow in contradiction with direct observations since statistically relevant spatial correlation exists at all scales in natural examples. Even if correlation models are still debated, the fractal model has been found to adequately represent correlation between natural fractures over scales [Bonnet *et al.*, 2001, and references therein]. This property renders that the fracture density (i.e., the number of fractures per unit volume) is generally decreasing with scale, the fractal dimension being related to the scale exponent that describes this scaling (see next section). The fractal model, that implies long-range correlations at every scale, makes the application of classical homogenization technique questionable if not irrelevant. Note that fractal is a space property that is irrelevant to describe fracture-size distribu-

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tions for instance. A complete discussion on the parameters necessary to statistically describe fracture network geometry is given in the next section.

[4] Spatial correlation is likely to change the connectivity rules of networks. For instance, it increases fracture clustering and thus the fracture-to-fracture connectivity. On the opposite, a heterogeneous fracture distribution also produces areas of low connectivity. At the system scale, both effects are competing, and the consequences on the whole-network connectivity are not trivial.

[5] The consequences of fractal correlation on fracture networks have never been studied, either on connectivity or on flow field. Several authors start with the assumption that the fractal correlation applies either to the permeability field [Sahimi and Mukhopadhyay, 1996], or to the flow field [Barker, 1988; Acuna and Yortsos, 1995]. However, the correlation models they used do not necessarily render the basic correlation of the fracture network structure. Moreover unlike the fracture network, the fractal dimension of permeability or flow field has not really been quantified in field experiments.

[6] In this paper, we aim at characterizing the connectivity properties of stochastic fractal networks. We first give the description of the geometrical model and of the numerical method developed to produce synthetic fractal fracture networks having any fractal dimension. We then characterize connectivity properties of networks made of constant, but small, fracture length. This simple case, close to classical assumptions made in percolation theory, should help to understand the very role of fractal correlation. We next deal with both fractal correlation and length distribution by treating the general fracture distribution model of [Bour et al., 2002] which assumes both a power law for fracture-length distribution and fractal correlation for fracture centers.

2. Preliminary Statements

2.1. Statistical Model of Fracture Network

[7] A geometric model of fracture networks involves a statistical description of the basic parameters of fractures: density, length, orientation, eccentricity, nature of intersections, etc. . . The role of some of these parameters on connectivity has been addressed in many studies amongst which [Robinson, 1983, 1984; Balberg et al., 1984; Berkowitz, 1995; Bour and Davy, 1997; de Dreuzay et al., 2000]. We let the reader refer to these studies for further information. Here we especially focus on parameters that have a scaling dependency, i.e., length and density.

[8] The fracture model that we will use is a basic first-order model initially proposed by Davy et al. [1990] and recently validated by Bour et al. [2002] over a fracture network mapped from the metric scale up to almost the kilometric scale in the Hornelen basin (Norway). The model gives the number of fractures centers, $n(l, L)dl$, observed within an elementary volume of typical size L , and whose lengths belong to the interval l and $l + dl$:

$$n(l, L)dl = \alpha L^D l^{-a} dl \quad \text{for } l \in [l_{\min}, l_{\max}]. \quad (1)$$

a is the exponent of the power law length distribution, D is the fractal dimension, α a constant. The only intrinsic characteristic length scales of the model are the smallest and

largest fracture length, $l_{\min}$ and $l_{\max}$ respectively. L is the scale at which the medium is generated for numerical simulations. In the field, it corresponds to the scale where a prediction is required, or where an experiment is designed. For numerical simulations, the system size L is chosen in between fracture length bounds, such as $l_{\min} \ll L \ll l_{\max}$ as it is for field conditions.

[9] This model can be easily extended to incorporate a fracture-orientation distribution such as: $n(l, L, \theta) dl d\theta = \alpha L^D l^{-a} \Psi(\theta) dl d\theta$. The orientation distribution will control percolation thresholds as shown by Robinson [1983, 1984] but not the scaling properties that are the main topics of this paper.

[10] A main issue, which is not addressed by this first-order model, is potential correlation between model parameters, for instance fracture position and length. These correlation that were reported in few studies, either from fault network analysis [Ackermann and Schlische, 1997; Bour and Davy, 1999; Darcel et al., 2003b], from hierarchical model [Clemo and Smith, 1997], or from mechanical arguments related to the description of fault-to-fault interactions [Renshaw and Pollard, 1994], generally renders a clustering organization that depends on fracture size. The role of such second-order correlation of the fracture model is potentially important, and will be addressed in a further study.

[11] Here we limit the study to the consequences of the scaling model described by equation (1) on connectivity. We consider idealized networks made of straight fractures generated in a 2-D plane with a random orientation distribution. Fracture-center positions are generated according to a fractal probability field (see section 2.4), and fracture lengths are drawn from a power law probability distribution, independently of the fracture position. All fractures have their center within the studied system (see Bour and Davy [1998] for an assessment of the role of fractures having their center outside the system while intersecting the studied system).

2.2. Parameters of the Statistical Model

[12] The exponent a , restricted to the range $[1, \infty]$, fixes the respective proportion of large and small fractures. The smallest is a , the largest is the proportion of large fractures. a lower than 1 is geologically irrelevant since this predicts a total fracture number that depends on the largest ones [Bonnet et al., 2001]. On the other end, if $a \rightarrow \infty$, all fractures have the same constant length $l_{\min}$. The fractal dimension D is kept within the range $[1, 2]$ for 2-D networks. If $D = 1$, the network has the dimension of a straight line. When $D = 2$ the fracture network model is no more fractal strictly speaking, but randomly, uniformly or regularly distributed over the 2-D plane.

[13] Compiling many field studies, Bonnet et al. [2001] or Renshaw [1999] reported power law exponents a for 2-D fracture maps in the range 1.5 to 3.5. The fractal dimensions are found to vary between 1 and 2; but most values are restricted within the range 1.5 to 2 [Bonnet et al., 2001].

[14] Another key parameter is the fracture density that is related to the total number of fractures, $N(L)$, present in a system of linear size L centered on the fracture network:

$$N(L) = \int_{l_{\min}}^{l_{\max}} n(l, L) dl = \frac{\alpha l_{\min}^{-a+1}}{(a-1)} L^D \quad (2)$$

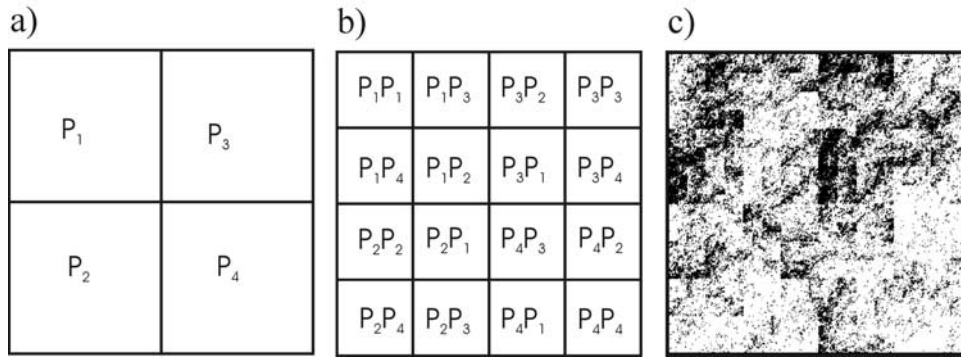


Figure 1. (a and b) First and second iterations of the multifractal field generation. (c) Example of a set of fracture positions obtained from a multifractal probability field with nine successive iterations.

Two density parameters can be defined. First, the apparent density, that is the number of fractures per unit area, is equal to $\gamma_2 = N(L)/L^2$. It is straightforward that the apparent density of the fractal decreases as L^{D-2} when increasing the system size, L . In contrast the fractal density that we define as $\gamma_D = N(L)/L^D$ remains constant whatever the system size. It thus can be considered as a robust—i.e., scale-independent—characteristic of the fractal network. The fracture density term α in (1), that is proportional to γ_D , fixes the average number of fracture per unit “fractal” area [Bour et al., 2002].

[15] Note that other definitions of fracture density are used in the literature, such as the mass (cumulative length) of fractures per unit area, or the normalized excluded area [Renshaw, 1999]. All these definitions are related to the different moments k of the distribution $n(l, L)$: $M_k(L) = \int_{l_{\min}}^{l_{\max}} n(l, L) \cdot l_c^k \cdot dl$, where l_c is the fracture length included into the system of size L [Bour and Davy, 1997]. The second moment of the distribution ($k=2$) is found to be a good proxy for connectivity [Robinson, 1983; Bour and Davy, 1997; Renshaw, 1999].

2.3. Definition and Measure of the Fractal Dimension

[16] Theoretically, the sole knowledge of D is not sufficient to fully characterize the fractal geometry of fracture networks. Indeed, a fractal object is characterized by an infinite set of generalized dimensions D_q , with $D_0 \geq D_1 \geq D_2 \geq \dots \geq D_\infty$, called the multifractal spectrum [Hentschel and Procaccia, 1983; Feder, 1988]. By definition, the multifractal spectrum is related to a measure (here the local fracture density) on a support (the object itself). The multifractal spectrum fully characterizes its level of regularity and homogeneity. The multifractal spectrum describes the behavior of the global quantities that is statistical averaged values of the measure on the geometrical support. The dimension $D_{q=0}$ of the multifractal spectrum is the dimension of the physical support, which can be either fractal or not. If $D_q = D_0$ whatever q , the measure is uniform on the support, and the system is a monofractal. If not, the system is said multifractal.

[17] The fractal dimension D as defined in (1) is the mass dimension of the fracture center distribution, and corresponds to the second-order dimension D_2 of the multifractal spectrum, also called correlation dimension D_c [Hentschel and Procaccia, 1983]. Indeed D_2 is defined by the scale

dependency of the density-density or pair correlation function $C_2(r)$, which is the number of points whose mutual distance is less or equal to r [Viseck, 1992]:

$$C_2(r) = \lim_{N \rightarrow \infty} \left(\frac{1}{N_2} \text{card} \{ (i, j); |x_i - x_j| \leq r, i, j \leq N \} \right)$$

$$D_2 = \lim_{r \rightarrow 0} \frac{\log C_2(r)}{\log r}$$

where N is the total number of points. Mathematically, this statistics over the pairs of points is equivalent to the average number of points included in a disk of radius r whose center is on the object. This is exactly the meaning of equation (1).

2.4. Network Generation

[18] To generate a fractal spatial density distribution of fractures, we have implemented a method based on a multiplicative cascade process [Schertzer and Lovejoy, 1987; Meakin, 1991]. The principle of the multiplicative cascade process is a recursive operation of fragmentation of a generic shape in n subdomains. Then a probability P_i , with $i = 1..n$, is attributed to each subdomain (Figures 1a and 1b). The fragmentation cascade can be iterated at infinity (practically set to 9 iterations); at each stage of the fragmentation process, a fragment takes one of the n probabilities P_i multiplied by the probability of the parent domain.

[19] The set of probabilities P_i defines the multifractal spectrum of the generated probability field. In theory, if l is the scale ratio of the elementary fragmentation process (i.e., ratio between the size of the parent domain and of each fragmented subdomains, equal to 2 for the example of Figure 1), the dimensions D_q of the eventual probability field are related to the set $\{P_i\}_{i=1..n}$ by:

$$\sum_{i=1}^n \frac{P_i^q}{(1/l)^{(q-1)D_q}} = 1 \quad (3)$$

Note that, for $q = 1$, equation (3) trivially expresses that the sum of a set of probabilities must be equal to 1. The method is flexible and allows us to generate a large range of fractal fields, from mono-fractal to multifractal spectrum.

[20] We choose a model with a fragmentation in 4 blocks of identical size ($l = 2$ in (3)). We fix the dimension D_2 ($q = 2$ in (3)) as the main characteristic of the generated probability

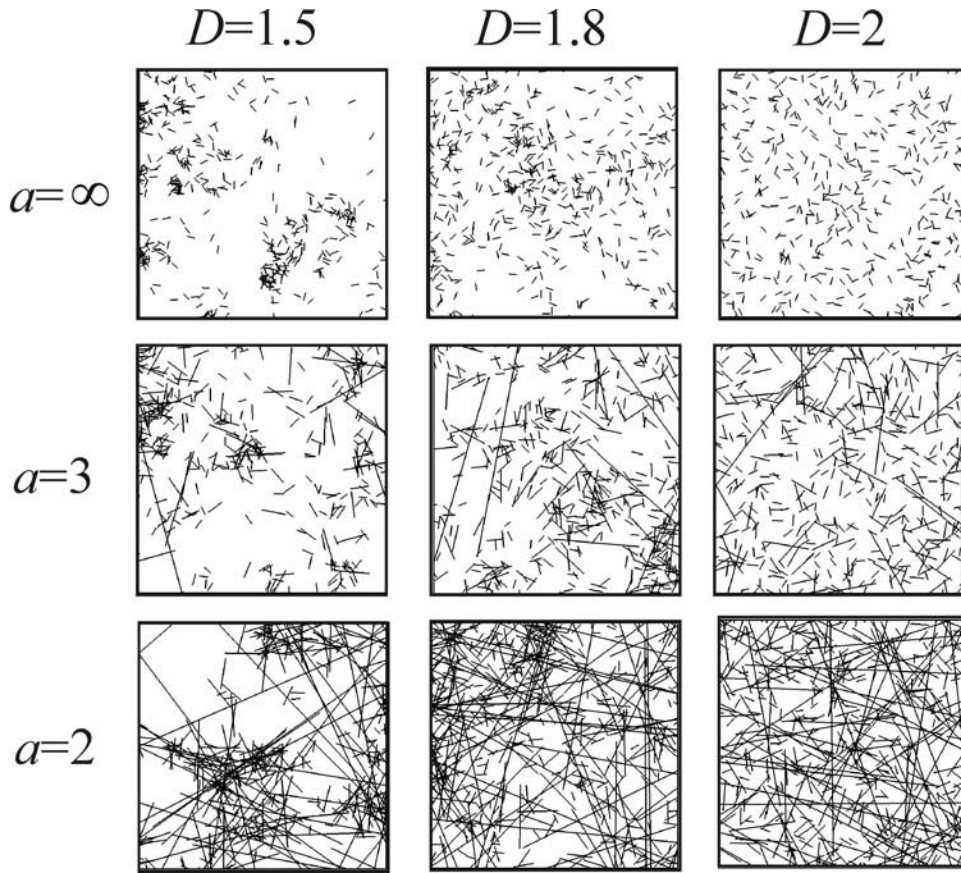


Figure 2. Examples of simulated fracture networks for several couples of parameters (a, D) . From left to right, the fractal dimension increases, reducing the clustering, up to $D = 2$, which corresponds to a uniform, noncorrelated field. From top to bottom, the proportion of large fractures compared to smaller ones increases, corresponding to a decrease of the power law length exponent a .

field, but we let the other dimensions $\{D_q\}_{q>2}$ undetermined, which means that they can take whatever value between 1 and 2. Also we choose an isotropic fragmentation process, meaning that the probabilities $\{P_i\}_{i=1,4}$ are randomly permuted before being mapped on the subdomains. Once the fractal probability field is generated, fracture centers are stochastically drawn from the probability distribution. An example of a set of points generated from a probability field at $D = 1.5$, and for nine cascades is given in Figure 1c. Figure 2 displays several synthetic fracture networks for different values of a and D . The smaller is D , the higher is the clustering, and the larger are empty areas. Note that in the stochastic fractal networks that we generate, no location is completely forbidden, so that the notion of “empty areas” refers to low probability areas that may nevertheless be eventually filled by fractures if the fracture density is increased enough.

[21] In practice, the multiplicative cascade process is iterated over a sufficient range so that the fractal structure is defined from a minimal scale, equal to the smallest fracture length $l_{\min}$, up to the fracture system size. Note that below this minimal scale the density distribution remains homogeneous whatever the fractal dimension. Therefore both scaling laws, related to density (i.e., to D) and to fracture length distribution (i.e., to a), are generated over the same range of scales. In the following, fracture networks are studied at percolation threshold. In practice

fractures are added to system up to the percolation threshold (connected pathway between opposite sides of the system) is reached.

2.5. Synthesis of Previous Connectivity Results

[22] Up to now connectivity properties have been only studied for fracture networks randomly distributed, i.e., for $D = 2$ in (1). First we briefly recall the main results obtained in that case. *Robinson* [1983] showed that systems made up of uniformly distributed cracks of constant length belong to the general framework of percolation theory [*Stauffer and Aharony*, 1992]. Percolation theory is a statistically based approach, which relates the physical properties of a system (connectivity, permeability, etc.) to an average geometrical measure, p called the parameter of percolation. For 2-D networks made up of cracks of constant length l , the parameter of percolation p is:

$$p = \frac{Nl^2}{L^2} \quad (4)$$

where N is the number of cracks of length l in a system of area L^2 . In this particular case, p turns out to be equivalent to the number of elements per unit area times a correction factor, which is the size of the element to the square. For infinitely large systems, there exists a critical value of the parameter of percolation p_c called percolation threshold

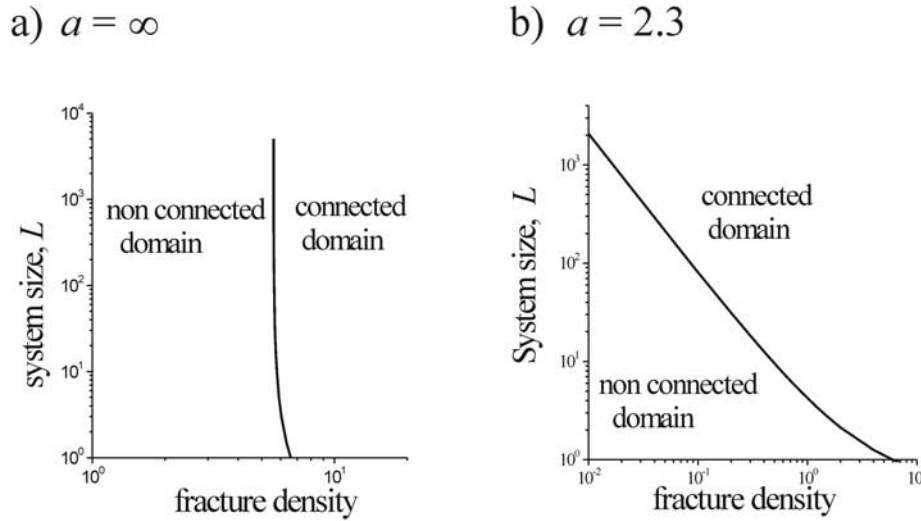


Figure 3. Synthesis of the connectivity regimes obtained (a) in the constant fracture length case and (b) in the widely scattered power law fracture length case ($a < 3$) for homogeneous networks, $D = 2$ (modified from *Bour and Davy* [1997]). Graphs represent the system size as a function of the fracture density γ for networks at the percolation threshold. At a given system size, i.e., on horizontal lines, systems are disconnected if the fracture density is smaller than the critical fracture density. Systems are thus disconnected on the left of the curves and, conversely, connected on the right of the curves.

below which systems are not connected, and above which systems are always connected.

[23] When dealing with atypical fracture geometry (for instance a widely scattered length distribution), the challenge is to find a parameter of percolation that is a right measure of network connectivity, i.e., that does not depend on scale at connectivity threshold. For 2-D random networks having a power law fracture length distribution the right expression of the percolation parameter is [*Bour and Davy*, 1997]:

$$p(L) = \int_{l_{\min}}^L \frac{n(l, L)l^2}{L^2} dl + \int_L^{l_{\max}} n(l, L) dl, \quad (5)$$

where L is the system size, $n(l, L)$ is the distribution given by (1), $l_{\min}$ and $l_{\max}$ are the lower and upper bounds of the fracture length distribution, respectively. The first term of (5) corresponds to the classical excluded volume concept [*Balberg et al.*, 1984], and the latter counts the number of fractures that are larger than the system size and that ensure the system connectivity by themselves. This expression has been proven to be very general and it can be extended to 3-D square or elliptic fracture networks [*Bour and Davy*, 1998; *de Dreuzy et al.*, 2000].

[24] In the case of a power law length distribution, two connectivity regimes have been characterized, depending on the ratio of fractures larger and smaller than the system size, conditioning the relative weight of both terms in (5) (Figure 3). In the first case, the classical case described by percolation theory, the percolation threshold corresponds to a constant density of fractures γ , apart from minor finite-size effects at small L (Figure 3a). This case is valid whatever the power law length distribution characterized by $a > 3$. In contrast, for an exponent a smaller than 3 (in 2-D), the connectivity increases with the system size (Figure 3b for

$a = 2.3$). In this regime, systems are on average disconnected at small scale, and connected at large ones. This result expresses the increasing control on connectivity by large fractures spanning the whole system, whose probability of occurrence increases with L . For $2 < a < 3$, both small and large fractures, expressed by the two terms of (5), rule the connectivity, whereas for $a < 2$ only large fractures, corresponding to the second term of (5), rule the connectivity. A more detailed discussion of these effects is given by *Bour and Davy* [1997, 1998] and *de Dreuzy et al.* [2000]. In the following these results are used as a reference to investigate the coupled effect on connectivity of the power law fracture length distribution, and of fractal correlations.

[25] The expression of the percolation parameter with fractal correlations is not straightforward. [*Berkowitz et al.*, 2000] proposed the following expression for constant-length fractures,

$$p_D = \frac{Nl^D}{L^D}, \quad (6)$$

with l the fracture length, extending the concept of excluded volume to a fractal domain whose size is L^D by definition. Since p_D is constant at the percolation threshold, (6) would mean that the number of elements, $N_c(L)$, necessary to reach the threshold scales as L^D .

3. Connectivity Properties of 2-D Fractal Networks Made Up of Small Fractures of Constant Length

3.1. Fracture Density at Percolation Threshold

[26] Here we attempt to define a percolation parameter capable to describe the connectivity state of any stochastic fractal fracture networks. For that purpose we first treat the

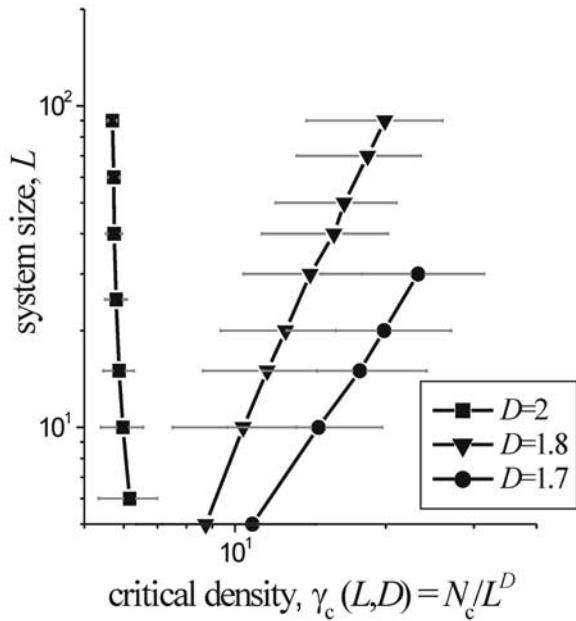


Figure 4. Same diagram as in Figure 3. Mean and standard deviation of the critical fracture fractal density at threshold for $D = 2$, $D = 1.8$, and $D = 1.7$.

basic case of 2-D fractal fracture networks where all fractures have the same length $l = l_{\min}$ to determine the sole effect of the fractal correlation. We choose $l_{\min} = 1$ for simplicity and without loss of generality. We performed numerical computations of the mean geometrical parameters of networks at percolation threshold. The evolution of the corresponding critical fractal density, $\gamma_c(L) = N_c(L)/L^D$, is plotted in Figure 4 for several values of the fractal dimension (D) and at several system sizes (L).

[27] It appears firstly that the nature of the percolation transition changes whether networks are fractal or not. Indeed for fractal fracture networks ($D < 2$), the transition width (standard deviation of $\gamma_c(L)$) remains large and finite for large systems whereas in the homogeneous case ($D = 2$) it vanishes when increasing the system size as predicted by percolation theory [Robinson, 1983; Balberg et al., 1984; Stauffer and Aharony, 1992]. The finiteness of percolation width was already observed by Prakash et al. [1992] and Schmittbuhl et al. [1993] as a consequence of long-range correlations in system structure. This behavior implies that the transition at threshold is no more a second order phase transition and suggests that percolation theory is not applicable for describing connectivity properties of correlated fracture networks. Secondly we notice that for the fractal fracture networks ($D < 2$), the critical fractal density at the percolation threshold increases with system size at a rate that depends on D , whereas for homogeneous fracture networks ($D = 2$) the critical density remains constant at the percolation threshold. By analyzing directly the evolution of the average number of fractures, $N_c(L)$, necessary to reach the percolation threshold over the finite range of scales investigated, we observe that it may be roughly approximated by a power law:

$$N_c(L) \sim L^X \quad (7)$$

with X larger than 2 as long as $D < 2$, and increasing while D decreases. It ensues directly that neither (4) nor (6) define a parameter that is scale invariant at percolation threshold. Moreover, the parameter defined in (6), which takes explicitly account of the fractal dimension, is even worse than (4) to define percolation threshold (because X is larger than 2).

[28] This result is in contradiction with the intuitive idea, expressed by (6) that, as the fractal is tight in a space $L^D < L^2$, the connectivity of a fractal system should be reached for a fracture number smaller than in the homogeneous case. If this assessment could be valid for some deterministic fractal like the Sierpinski carpet, it is no more appropriate for the stochastic fractals that we generate since it neglects the counterpart of the fractal clustering, that are lacunarity between clusters. This effect is illustrated in Figure 5, where synthetic fracture networks at the percolation threshold are displayed along with their corresponding infinite clusters (thick elements). Many fractures of the fractal networks (Figure 5 right column) to be compared with the uniform distribution on the left column) concentrate in clusters without improving the global connectivity of system. Conversely the large-scale connectivity appears to be ruled by the occurrence of a few fractures in low-probability zones, so that the large scale connectivity is controlled by the spatial repartition of the low-density areas. In fractals, the proportion of these critical low-probability zones increases with system size entailing that the density of fractures at threshold must increase just to maintain a constant number of fractures in low-probability zones. This explains why fractal networks are disconnected at large scale if the fractal density is kept constant (Figure 4), and why for a given fractal density, the higher is the clustering (small D), the smaller is the critical system size at which the fracture network appears disconnected.

[29] To summarize, the study of the critical density of fractures at percolation threshold shows that, for stochastic fractal fracture networks made of constant length fractures, we cannot validate a simple, and sound, expression of the percolation parameter that is constant (i.e., scale-invariant) at connectivity threshold. Moreover, the fast increase with scale of $N_c(L)$, the number of fractures at threshold, makes difficult direct computations of threshold at large scales.

3.2. Definition of a Local Parameter of Percolation, and Application

[30] From the previous part, it is clear that for fractal networks made of small constant-length fractures ($a \rightarrow \infty$), neither the mean fracture density nor the mean fractal fracture density can represent the connectivity state of stochastic fractal fracture networks. For a fixed number of elements, the number of intersections between fractures increases with fractal clustering, without increasing directly the connectivity at a scale larger than the clusters sizes.

[31] In most of classical percolation systems, the threshold is reached when the space occupied by connected objects reaches a critical proportion of the total available space. This concept, which leads to a critical occupation probability for regular lattices, is also basic to the excluded-volume theory defined for a population of random cracks. We will show that it can be also applied to fractal fracture network, providing that 1) the total space is divided into elementary volumes, and 2) that a local density criterion is

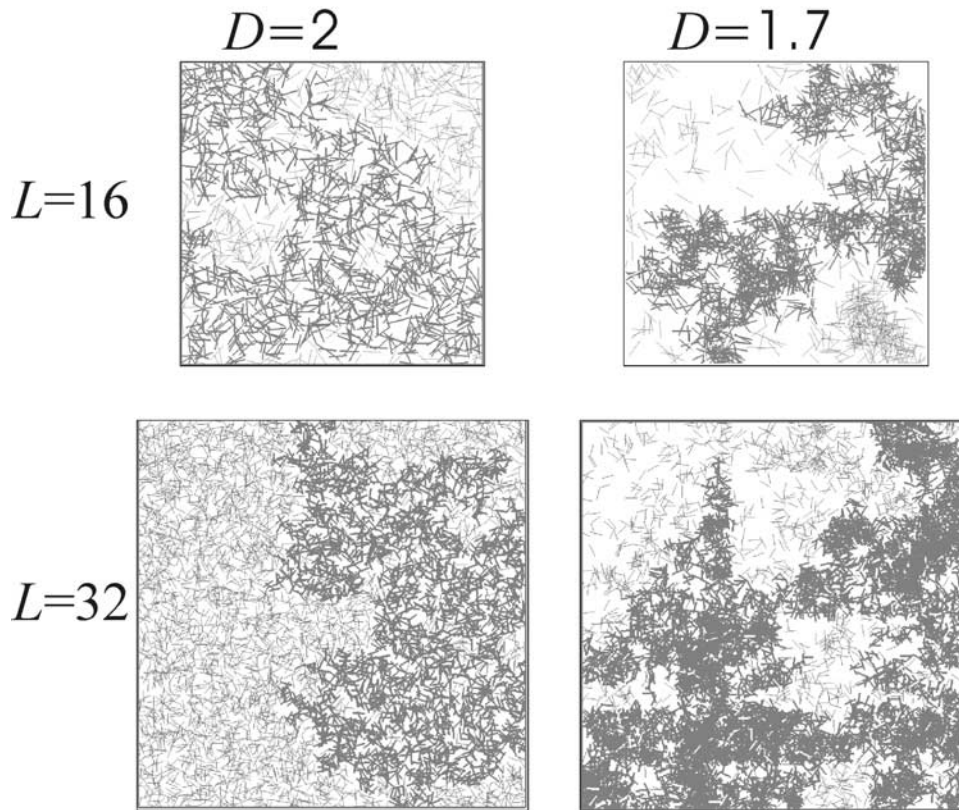


Figure 5. Fracture networks generated at the percolation threshold for system sizes increasing from (top) $L = 16$ to (bottom) $L = 32$. On the left side, the fracture center distribution is homogeneous ($D = 2$), whereas on the right side it is fractal with $D = 1.7$. Dark fractures form the connected cluster whereas light fractures form the isolated clusters. For $D = 1.7$, a large amount of fractures concentrates in high-probability zones forming dense clusters without improving the system connectivity, whereas the system connectivity is controlled by the occupation of the low-probability zones.

applied to define the connectivity state of each volume. If the scale of elementary volumes and the local connectivity criterion are properly defined, the connectivity is reached for a fixed percentage of elementary volumes.

[32] A natural scale for this coarse-grained averaging is the fracture length $l_{\min}$ since it is the scale below which the spatial correlation does not play any role on the fracture-to-fracture connectivity. The averaging procedure is illustrated in Figure 6, where first, a grid of mesh $l_{\min}$ is applied onto a fracture network (Figure 6a) to enable, next, the computation of the local fracture density distribution (Figure 6b), the local density being proportional to the cumulative length (mass) included in a mesh. The following step consists in defining a local connectivity criterion α that statistically determines if a mesh is connected or not to its neighbors. α is a local-density criterion similar to those defined for random fractures [Robinson, 1983], while $n_c(\alpha)$, the proportion of meshes whose local density is larger than α , is a site-type criterion. If α increases, n_c decreases; $n_c(\alpha)$ gives a cumulative measure of the local density distribution at percolation threshold. This “percolation” curve may change with system size or fractal dimension (Figure 7).

[33] In the homogeneous case ($D = 2$), the distribution of the local densities is simply a Gaussian whose mean value is equal to 5.6 at percolation threshold (5.6 being the value

of the percolation parameter at threshold for $l_{\min} = 1$ [Robinson, 1983]). This fixes the curve $n_c(\alpha)$ (Figure 6b), which predicts that 50% of the meshes are considered connected if $\alpha = 5.6$. Note that, in this case, α is both a local and a global density criterion, because of the uniform-density assumption. Figure 6b shows that changing the system size does not modify the $n_c(\alpha)$ curves, and thus the density distributions at threshold, except for small finite-size effects described by the classical percolation theory [Stauffer and Aharony, 1992].

[34] This coarse-grained averaging is now applied to the correlated fracture networks made of constant length fractures at percolation threshold. Results from Monte Carlo simulations are presented in Figure 7, for different values of the fractal dimension D at different system sizes. The fractal dimension clearly controls the broadness of the local density distribution, and thus the shape of the curve $n_c(\alpha)$ (Figure 7a). For a given system size, the lower is the fractal dimension, the broader is the density distribution, and the flatter is the $n_c(\alpha)$ curve. The most important result is that all curves intersect at about $\alpha \approx 5-6$ and $n(\alpha) \approx 0.5$ whatever the fractal dimension (Figure 7a) and the system size (Figure 7b), meaning that having 50% of the system connected at the local scale leads to the connection at the system scale, whatever the fractal dimension. The invariance of this $(\alpha, n_c(\alpha))$ point with respect to model param-

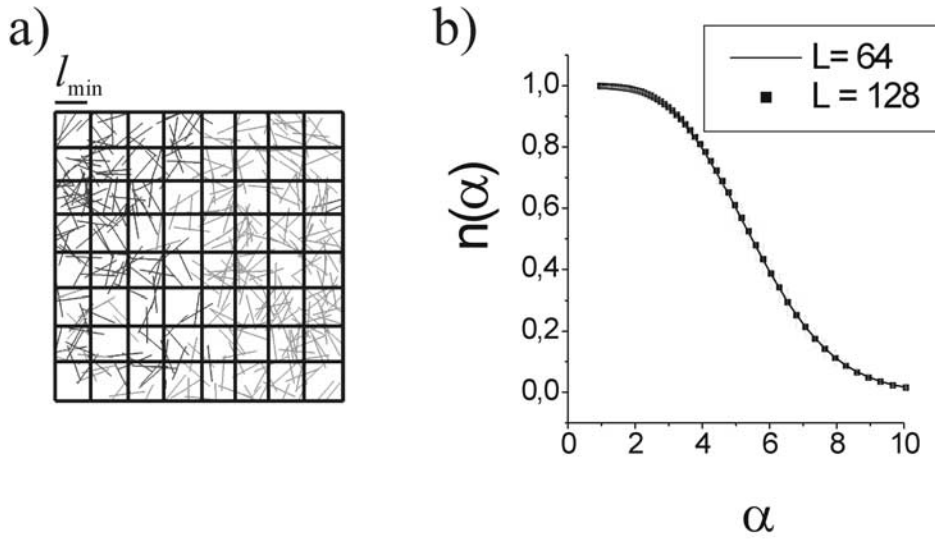


Figure 6. Method for computing the fracture density distribution. (a) A uniform fracture network at the percolation threshold for $L = 8$, and for a constant fracture length $l = l_{\min} = 1$. A grid of mesh size equal to $l_{\min}$ is superposed on the network. (b) Proportion of meshes, $n(\alpha)$, whose fracture density is larger or equal to α for $D = 2$ and for two system sizes.

eters seems to demonstrate the validity of the coarse-graining procedure, and the way it can be used to define the percolation threshold. This result makes also possible the direct prediction of the critical density of fractures at threshold straightly from the spatial probability field. This can avoid tedious intensive computations. By applying the criterion $n(\alpha = 5) = 0.5$, we found a good agreement with the computed fracture density at percolation threshold (Figure 4).

[35] Finally the curves $(\alpha, n_c(\alpha))$ provide a thorough explanation of the evolution of the critical fracture density at the percolation threshold for stochastic fractal fracture networks (Figures 4 and 7). Indeed, although the proportion of meshes to be filled at threshold is constant ($n(\alpha_c) = 0.5$), the contrast between areas of high and low probabilities increases when L increases (or D decreases), leading to an increase of the critical density quantified by the thickening of the tail of $n_c(\alpha)$.

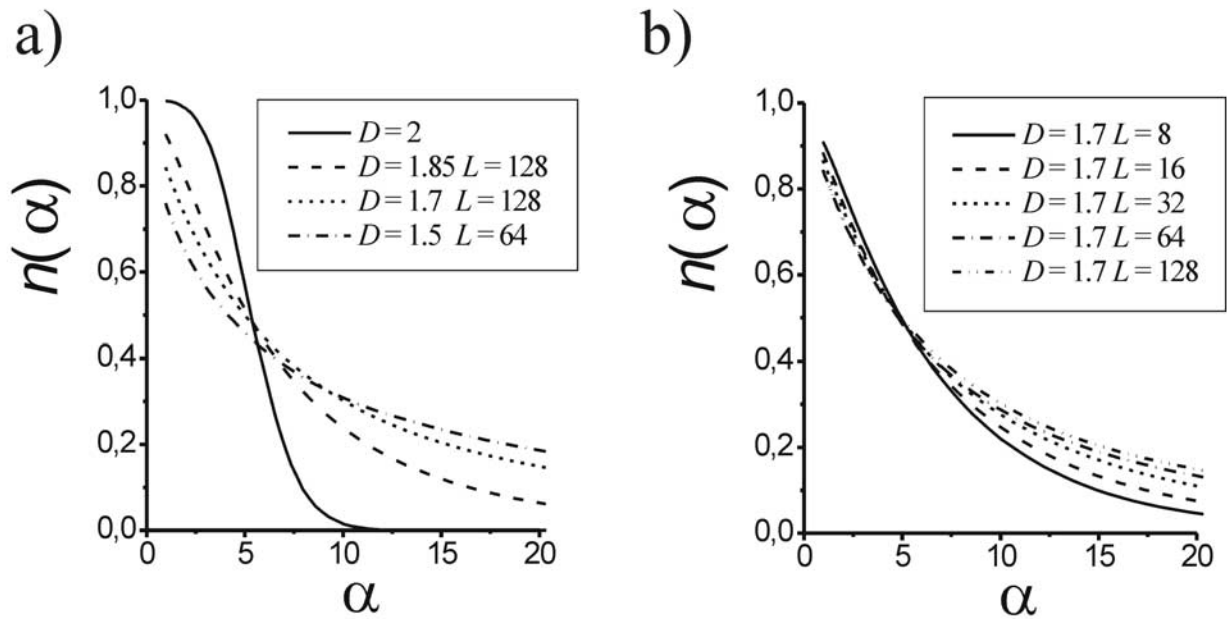


Figure 7. (a) Curves $(\alpha, n(\alpha))$ at percolation threshold for fracture networks generated for different values of the fractal dimension. (b) Curves $(\alpha, n(\alpha))$ at percolation threshold for a constant fractal dimension ($D = 1.7$) at different system sizes.

[36] In summary, the results obtained on constant-length fracture networks show that the fractal property creates both clusters and anticlusters, and that anticlusters (i.e., holes or low-density areas) control the large-scale connectivity. As a consequence, for a given fractal density, stochastic fractal fracture networks can hardly be connected and there exists a critical scale at which they are surely disconnected. We do recognize nevertheless that further work is necessary to completely assess the existence of such an invariant critical point at percolation threshold. Indeed if the invariance with L seems rather robust, we notice some variations with the multifractal spectrum. Such a complementary study relates to the models studied by [Meakin, 1991; Sahimi and Mukhopadhyay, 1996; Marrink et al., 2000], who analyzed the percolation transition on regular lattices with long-range correlations. It will be the subject of a separate work.

4. Connectivity of 2-D Fractal Networks With a Power Law Length Distribution

4.1. Expected Behavior at Threshold

[37] The introduction of a widely scattered length distribution in the system leads to another kind of long-range correlation coming directly from fracture lengths. Indeed, a fracture induces by itself a potential connectivity over a scale that corresponds to its length. This effect can be dramatic if the probability to find fractures as large as the system size is not negligible. The power law length distribution makes possible this configuration. For instance, if $a < 2$, the system is essentially connected by large fractures that span the whole system [Bour and Davy, 1997]. For such small values of the exponent a , we do not expect fracture clustering to be relevant for assessing the global connectivity properties. For the complete fracture model with both a power law length distribution and fractal correlations, the issue is thus to know which long-range correlation will be dominant, and to define the transition between a regime controlled by the length distribution (when the exponent a is small enough), and a regime controlled by the fractal correlations (when the exponent a is large enough, as it is for the constant-length model where $a = \infty$). In the following, we thus investigate the ranges of exponents a and D to characterize the behavior of networks at percolation.

[38] We first consider the number of large fractures within the system, because of their key role in connectivity. If α and β are two constant coefficients close to 1, the number of fractures whose length is in the range $[\alpha L, \beta L]$ is:

$$N(\alpha L < l < \beta L) = \int_{\alpha L}^{\beta L} n(l, L) dl \propto L^{D-a+1}, \quad (8)$$

Equation (8) first shows that the self-similar property, that is the scale-invariance of fracture patterns, is achieved if and only if $a = D + 1$ since the scale-invariance of the number of structures observable at a given scale is a necessary condition for the self-similar property. This also means that a fractal property does not necessarily mean self-similarity, contrarily to an intuitive opinion.

[39] If $a < D + 1$, $N(\alpha L < l < \beta L)$ increases with scale, leading to an increase of the number of large fractures, and

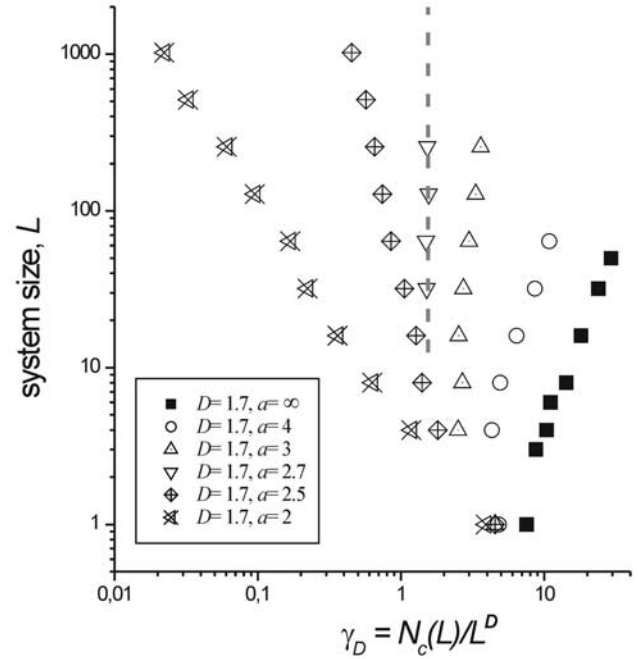


Figure 8. Plot of the critical scale at which percolation occurs, for different fractal density γ , for different values of a and for $D = 1.7$.

thus to a necessary increase of the connectivity with scale. Conversely if $a > D + 1$, the probability to observe large fractures decreases with scale, so that connectivity is eventually ruled, for infinitely large systems, by fractures that are much smaller than the system size. This case is thus similar to the constant-length model described above; and we thus expect a decrease of the connectivity with scale due to fractal correlation (i.e., if $D < 2$) as shown in the previous section.

[40] For $a = D + 1$, that corresponds to the self-similar case, the number of fractures whose length is of the order of L is independent of L . We thus expect the connectivity properties also to be scale-invariant. Indeed if the pattern is connected at a given scale, then it must be connected for larger systems. And if it is disconnected at a given scale, it must be disconnected for smaller systems.

[41] To verify these conjectures, we compute systems at percolation threshold for a large range of fracture densities, power law exponents and fractal dimensions. Results of these simulations confirm the different connectivity regimes expected. They are illustrated on Figure 8, for the case $D = 1.7$ and a broad range of a values. As long as $a > D + 1$, the critical density increases when increasing system size, as previously shown for the limit case of constant fracture lengths ($a = \infty$, solid squares). This trend is still obtained for $a = 3$ and $D = 1.7$ (up triangles), even if slightly less than for constant fracture lengths. For $a = D + 1$ ($D = 1.7$ and $a = 2.7$ here), the critical fractal density at threshold remains constant whatever system size. At last, for $a < D + 1$ (left triangles for $a = 2$, diamonds for $a = 2.5$) we notice a decrease of the critical density at percolation threshold when the system size increases, as it is in the homogeneous case ($D = 2$) when a is smaller than 3 [Bour and Davy, 1997]. Note that the variations with scale of the critical density at

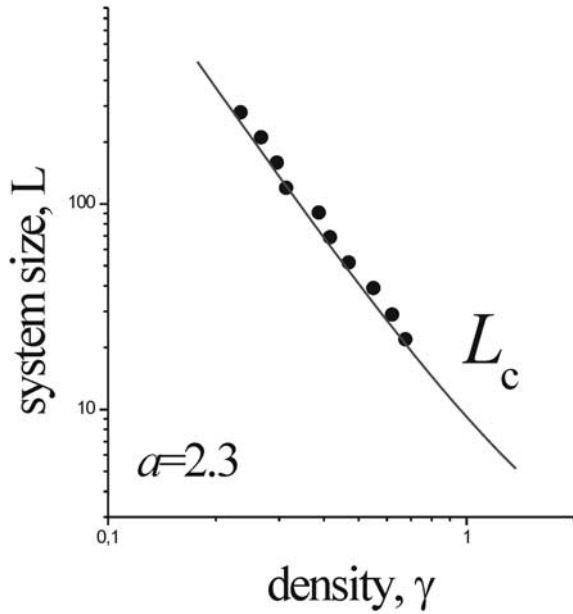


Figure 9. Critical system size at which percolation occurs, L_c , depending on the fracture density, γ for $a = 2.3$, and $D = 1.5$. The curve, which corresponds to theoretical predictions (equation 9), is in good agreement with the numerical results (solid circles). For a given fracture density, fracture systems are, on average, connected when considering systems having a size larger than the critical system size L_c .

threshold can be easily predicted for $a \leq D + 1$ (see Appendix A).

4.2. Critical Scale of Connectivity

[42] Additional simulations have also confirmed that for $a > D + 1$, the parameter of percolation at threshold $p_c(L)$ defined in (5) varies significantly with the system size L . This result is consistent with results obtained for $a = \infty$ (previous section). In that case, fractal correlations control system connectivity, and the “percolation” parameter defined by equation (5) is no more relevant to describe unambiguously percolation threshold. On the other hand, if $a \leq D + 1$, $p_c(L)$ is invariant at threshold, and can thus be used to quantify network connectivity. The width of the transition remains nevertheless finite even for large systems implying that the transition at threshold is not abrupt at large scale, as it is in classical percolation theory or in second-order phase transitions.

[43] In Appendix A, we calculate analytically the scaling of the average number of intersections per fracture for different values of a and D . This fracture intersection number is often used to describe the connectivity properties of fracture networks, and can be easily used to qualify natural examples. It is an invariant at percolation threshold for random fracture networks made of small finite length [Robinson, 1984; Hestir and Long, 1990; Berkowitz and Balberg, 1993]. For $a < D + 1$, the fracture intersection number at threshold, I_c , is no more a scale invariant; it decreases as a power law of scale with an exponent that depends on a , or on both a and D (see Appendix A and Figure A1). The scale invariance of I_c is recovered for

$a = D + 1$, in agreement with the self-similar nature of such network (see Appendix A). However, I_c depends on the fractal dimension D ; the lower is D , the larger is I_c (Figure A2). This latest result, that is consistent with an increase of fracture clustering as the fractal dimension decreases, may have strong implications on the transport properties of fractal fracture networks.

[44] Another way to check the preceding developments for $a \leq D + 1$, is to compute analytically the critical scale at which the percolation threshold is reached for a given fracture density (Figure 8). For $a < D + 1$, assuming the threshold equality $p(l, L) = p_c$, and combining (1) and (5), one obtains:

$$\frac{l_{\min}^{3-a}}{a-3} L_c^{D-2} + \frac{2}{(3-a)(a-1)} L_c^{D+1-a} = \frac{p_c}{\gamma_c(a-1)} l_{\min}^{D+1-a},$$

that provides L_c for large system sizes through:

$$L_c = \left[\frac{(3-a)p_c}{2} l_{\min}^{-a+D+1} \frac{1}{\gamma_c} \right]^{\frac{1}{D+1-a}}. \quad (9)$$

This expression is plotted in Figure 9 and compared with numerical simulations for $a = 2.3$. Numerical computations are in good agreement with analytical developments.

5. Discussion

[45] The previous developments entail different regimes of connectivity depending on the respective values of a and D , above, at and below the critical value $a = D + 1$. These results have strong implications for the connectivity properties of natural fracture networks. For a given network, i.e., a given fracture density, one expects the network to be better connected at small scale than at large scale if a is larger than $D + 1$. On the opposite, we expect the connectivity to increase with scale for $a < D + 1$. Interestingly, it is only for $a = D + 1$ that the connectivity depends only on fracture density.

[46] From data analyses of fracture trace maps, we expect the exponent a of the length distribution to vary within the range 1.5 to 3.5, with most values between 1.7 and 2.8 [see Bonnet et al., 2001, and references therein]. The values of the fractal dimension measured from fracture trace maps are widely scattered but most values are ranging between 1.5 and 2. For natural fracture systems, we thus expect $a \leq D + 1$ in most cases. As a consequence, the connectivity of fracture systems should either increase with the scale of observation or being invariant with scale. Note that the latter case, which corresponds to self-similar systems ($a = D + 1$), may be encountered, as shown by Bour et al. [2002], who found $a = 2.75$ and $D = 1.8$ for the Hornelen fracture network (Norway). Great care should be nevertheless taken when extrapolating our results to natural examples. In particular, in the previous developments we neglected some geometrical properties that may rule partly the connectivity properties of fracture network, like the local correlations between fractures and the nature of fracture intersections.

[47] Even if calculations were performed on 2-D fracture networks, we are quite confident that most of the results are still qualitatively valid in 3-D, as it is for power law length distribution [Bour and Davy, 1997, 1998]. In particular we

expect that $a = D + 1$ fixes a transition of connectivity regimes since in 3-D, as in 2-D, the relationship between a and $D + 1$ fixes the scaling of the proportion of large fractures with system size (which decreases if $a > D + 1$, and increases otherwise, see equation 8). In terms of fracture model parameters, which are generally derived from 2-D outcrops, Darcel *et al.* [2003a] recently showed that the simplest stereological rules, $a_{3-D} = a_{2-D} + 1$ and $D_{3-D} = D_{2-D} + 1$, apply as long as $a_{3-D} > 2$, the subscripts 2-D and 3-D referring to two- and three-dimensional systems respectively. From 2-D field data analysis [Bonnet *et al.*, 2001], we thus expect a_{3-D} to range within 2.7 and 3.8, while D_{3-D} should vary between 2.5 and 3. The case $a < D + 1$ is thus likely to be encountered for most natural cases either in 2-D or 3-D, and we should expect an increase of connectivity with scale. Note that percolation threshold are found to be much lower in 3-D than in 2-D for regular lattice networks [Stauffer and Aharony, 1992]; this effect which emphasizes an increase of the connection possibilities should also be found when comparing 2-D and 3-D fracture networks.

6. Conclusions

[48] In this paper, we enlarged previous study of the connectivity properties of fracture patterns by studying the consequences of fractal clustering. According to recent measures on natural fracture outcrops, we considered networks that have two basic geometrical properties: a fracture-center density that is fractal and a fracture length distribution that follows a power law l^{-a} . Both geometrical properties, that are characteristic of most natural fracture patterns, introduce long-range correlations that may be crucial for the network connectivity. Different regimes of connectivity are identified depending on the fractal dimension D , and on the length-distribution exponent a .

[49] For $a < D + 1$, system connectivity is dominated by the fracture length distribution, with a negligible contribution of fractal correlations. The system connectivity is mostly ruled by the largest fractures whose length is of the order of, or larger than, the system size. The connectivity increases with scale so that one expects fracture networks to be always connected at large scale. The number of intersections per fracture, which may partly control transport properties of network, is dependent on the fractal dimension D .

[50] For $a > D + 1$ the connectivity is ruled by fractures much smaller than the system size, with a strong control by spatial correlation. The connectivity now decreases with system scale. This means that large systems are always disconnected, even if it can be connected at finite scale.

[51] At last, the self-similar case ($a = D + 1$) corresponds to the transition between these two regimes. The connectivity properties were found scale-invariant, so that the percolation threshold corresponds to a critical fractal density. The average number of intersections per fracture at threshold is also a scale invariant.

Appendix A: Average Number of Intersections per Fracture

[52] The analytical expression of the average number of intersections per fracture, noted $I_c(L)$ (subscript c meaning at percolation threshold), may be derived knowing the

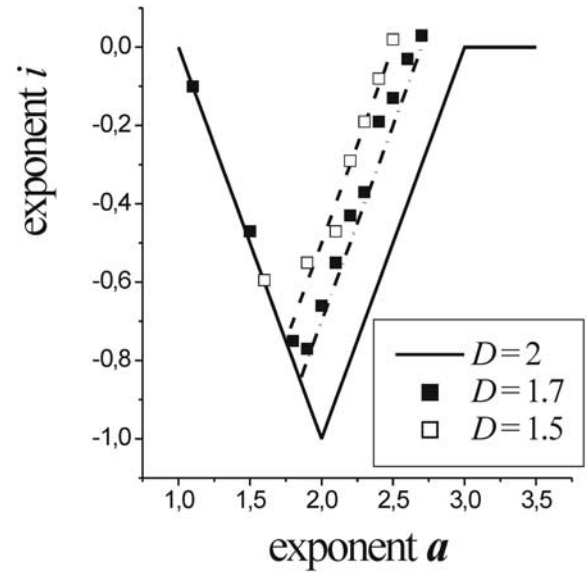


Figure A1. Exponent of the scale dependency of the average number of intersections per fracture at percolation threshold, $I_c(L) \sim L^i$, for different values of the fractal dimension. Straight lines correspond to theoretical predictions.

expression of the fracture length distribution at threshold, $n_c(l, L)$ and of the probability of intersection between two fractures of different lengths, $p(l, l')$ [Bour and Davy, 1997]. Indeed the sum over l' of the product $p(l, l') n_c(l', L)$ provides the average number of intersections per fracture of length l . This quantity may be then summed to obtain the total number of intersections in the system that provides $I_c(L)$ once divided by $N_c(L)$, the number of fractures at threshold.

[53] Combining the invariance of the percolation parameter at threshold, $p_c(L)$, and equation (5) implies that the critical number of fractures in the system at threshold, $N_c(L)$, is scaling with L such as:

$$N_c(L) \propto L^{a-1}, \text{ for } 1 < a \leq D + 1. \quad (\text{A1})$$

Note that this peculiar scaling obtained at threshold (A1) comes from scale effects that are inherent to systems where the connectivity is controlled by large fractures [Bour and Davy, 1997]. Note also that we do not expect scale effects for the self-similar case ($a = D + 1$); the critical density to reach the percolation threshold should be indeed independent of the system size $\gamma_c(L) = N_c(L)/L^D \sim L^0$. Since the expression of the critical number of fractures at threshold for different system sizes is determined clearly only for $a \leq D + 1$ (A1), we restrict our analysis for $I_c(L)$ to $a \leq D + 1$. For $a > D + 1$, $N_c(L)$ is anyway expected to diverge without the existence of an invariant so that $I_c(L)$ should also diverge.

[54] In another work [Darcel *et al.*, 2003a], we have shown that the intersection probability between two fractures of length l and l' , in a system of fractal dimension D , follows:

$$p(l, l' < l, L) \propto \frac{l' \cdot l^{D-1}}{L^D} \quad (\text{A2})$$

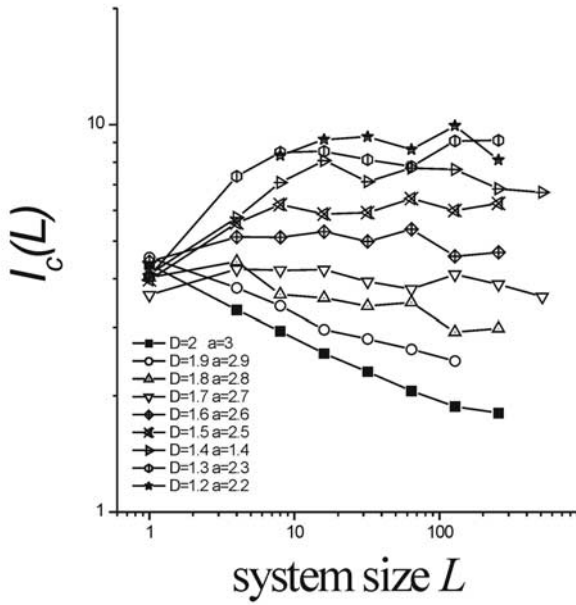


Figure A2. Computed values of the average number of intersections per fracture at percolation threshold as a function of the system size for the self-similar case $a = D + 1$.

Now, for $1 < a \leq D + 1$, combining (A1) and (A2) leads to:

$$I_c(L) \sim L^{1-a} \text{ for } 1 < a \leq (D + 2)/2$$

$$I_c(L) \sim L^{a-1-D} \text{ for } (D + 2)/2 \leq a \leq D + 1$$

[55] In Figure A1 we reported the scaling exponent of $I_c(L)$ at threshold, for $D = 1.5$ (open squares), $D = 1.7$ (solid squares) and $D = 2$. Results are in excellent agreement with theoretical predictions. For $1 < a \leq (D + 2)/2$, $I_c(L)$ is varying as the inverse of $N_c(L)$ meaning that the total number of intersections at threshold is scale independent. We indeed checked that the total number of intersections is extremely small whatever the system size, in agreement with the fact that fracture networks are always connected by fractures having a size equal or larger than the system size for such a range of values of a and D .

[56] For $(D + 2)/2 \leq a < D + 1$, the total number of intersections increases with system size, but not as fast as the number of fractures in the system so that $I_c(L)$ decreases with scale at threshold. Like for $1 < a \leq (D + 2)/2$, this scaling is consistent with the fact that the occurrence probability of fractures having a length of the order or larger than the system size is increasing with L .

[57] At last, $I_c(L)$ does not vary with L for $a = D + 1$ in agreement with the self-similar nature of such network. Numerical computations confirm this scale-invariance whatever the couple of values of a and D as far as $a = D + 1$ (Figure A2).

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